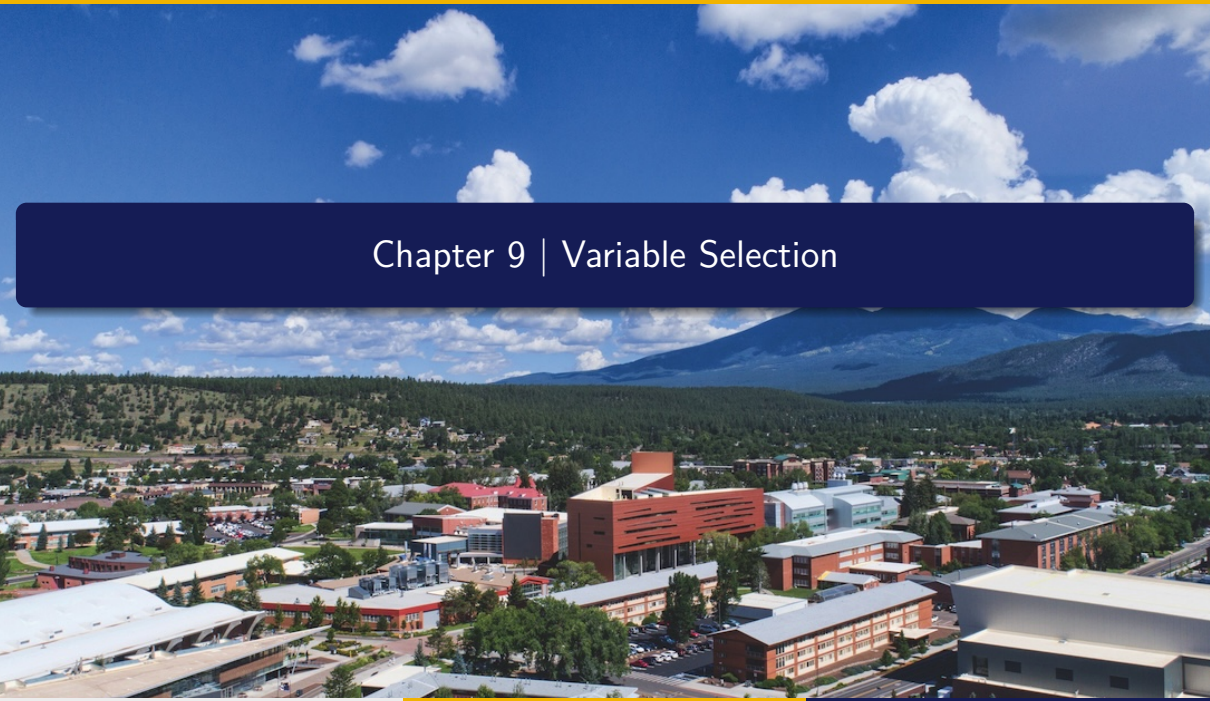
Chapter 9 | Variable Selection



Learning objectives

- Given a set of data, we want a simplest model that explains adequately the observed data patterns.
- Upon successful completion of this chapter, we will be able to do the following tasks:
  1. Find a best subset of the predictors using the three variable selection methods (backward elimination, forward selection, and stepwise selection).
  2. Find a best subset of the predictors using the model selection criteria (AIC and BIC).

Outline

1. Introduction
2. Nested Models
3. Testing-based Model Selection
4. Criterion-based Model Selection

Motivation

- Suppose a data set contains observations of the response variable Y and p predictors $X_1, \ldots, X_p$ from a sample of n subjects.

Subject	Y	X_1	$\cdots$	X_p
1	y_1	x_{11}	$\cdots$	x_{1p}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
n	y_n	x_{n1}	$\cdots$	x_{np}

- If we fit a linear model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ with large p (but still $p < n$), three issues can arise.

Potential issues

- Issue 1: **Multicollinearity**. If all predictors are used, it is very likely (almost always) that some predictors are co-linear (linearly dependent).
  1. Suppose $\mathbf{x}_1 = \begin{bmatrix}1 & 1 & 1 & 1\end{bmatrix}^\top$ and $\mathbf{x}_2 = \mathbf{x}_1 + 1/4000$ are predictors. Then,

$$\mathbf{X} = \begin{bmatrix}\mathbf{x}_1 & \mathbf{x}_2\end{bmatrix} = \begin{bmatrix}1 & 1.00025\\ 1 & 1.00025\\ 1 & 1.00025\\ 1 & 1.00025\end{bmatrix} \Rightarrow \mathbf{X}^\top\mathbf{X} = \begin{bmatrix}4.000 & 4.001\\ 4.001 & 4.002\end{bmatrix} \Rightarrow (\mathbf{X}^\top\mathbf{X})^{-1} = \begin{bmatrix}-4002000 & 4001000\\ 4001000 & -4000000\end{bmatrix}$$

  2. Now, suppose $\mathbf{x}_2 = \mathbf{x}_1 + 1/4000 + 1/8000000$. Then,

$$\mathbf{X} = \begin{bmatrix}\mathbf{x}_1 & \mathbf{x}_2\end{bmatrix} = \begin{bmatrix}1 & 1.000250125\\ 1 & 1.000250125\\ 1 & 1.000250125\\ 1 & 1.000250125\end{bmatrix} \Rightarrow \mathbf{X}^\top\mathbf{X} = \begin{bmatrix}4.000 & 4.001000\\ 4.001 & 4.002001\end{bmatrix} \Rightarrow (\mathbf{X}^\top\mathbf{X})^{-1} = \begin{bmatrix}1334000 & -1333667\\ -1333667 & 1333333\end{bmatrix}$$

  3. **Note**: $\mathbf{X}^\top\mathbf{X} \cong \mathbf{X}^\top\mathbf{X}$. However, $(\mathbf{X}^\top\mathbf{X})^{-1} \cong (-3)(\mathbf{X}^\top\mathbf{X})^{-1}$, implying small changes, perhaps caused by rounding, can give substantially different and unstable inverses when predictors are linearly dependent.

Potential issues

- Issue 1: **Multicollinearity** (continued)
  4. Compare the LS estimates:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y} = \begin{bmatrix} -4002000 & 4001000 \\ 4001000 & -4000000 \end{bmatrix}\mathbf{X}^\mathsf{T}\mathbf{y} \text{ vs. } \hat{\boldsymbol{\beta}} = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y} = \begin{bmatrix} 1334000 & -1333667 \\ -1333667 & 1333333 \end{bmatrix}\mathbf{X}^\mathsf{T}\mathbf{y}$$

  5. **Note**: $\hat{\boldsymbol{\beta}} \approx (-3)\hat{\boldsymbol{\beta}}$, implying the LS estimates are unreliable.
  6. Denote $\hat{\boldsymbol{\beta}} = \begin{bmatrix}\hat{\beta}_0 & \hat{\beta}_1\end{bmatrix}^\mathsf{T}$. Because $\text{SE}(\hat{\beta}_1) = \hat{\sigma}\sqrt{(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}_{[2,2]}}$ is very large, a t-test statistic for $\text{H}_0 : \beta_1 = 0$

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{\hat{\sigma}\sqrt{(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}_{[2,2]}}}$$

     will be very close to 0, producing a large p-value. Therefore, we do not reject $\text{H}_0 : \beta_1 = 0$.
  7. **Note**: If some predictors are strongly correlated (linearly dependent), $\text{SE}(\hat{\beta}_j)$ becomes large (because $(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}_{[j+1,j+1]}$ is large). Therefore, no regression coefficient estimates become significant.

Potential issues

- Issue 2: **Overestimation of σ^2**. Unnecessary predictors will waste degrees of freedom, possibly overestimating the model error variance σ^2.
  1. Consider a regression model with response variable **Y** and predictors $X_1, \ldots, X_5$:

     $$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_5 x_{i5} + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathsf{N}(0, \sigma^2), \quad i = 1, \ldots, 50$$

     The model error variance σ^2 is estimated by

     $$\hat{\sigma}^2_{X_{[,1:5]}} = \frac{1}{50-6} \sum_{i=1}^{50} (y_i - \hat{y}_i)^2, \quad \text{where } \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_5 x_{i5}$$

  2. Suppose additional predictors $X_6, \ldots, X_{10}$, which are not related with **Y**, are included:

     $$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_5 x_{i5} + \beta_6 x_{i6} + \cdots + \beta_{10} x_{i10} + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathsf{N}(0, \sigma^2), \quad i = 1, \ldots, 50$$

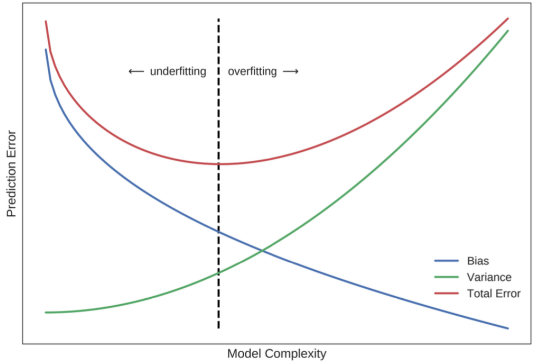
     Then,

     $$\hat{\sigma}^2_{X_{[,1:10]}} = \frac{1}{50-11} \sum_{i=1}^{50} (y_i - \hat{y}_i)^2, \quad \text{where } \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_5 x_{i5} + \hat{\beta}_6 x_{i6} + \cdots + \hat{\beta}_{10} x_{i10}$$

  3. **Note**: $\hat{\sigma}^2_{X_{[,1:5]}} < \hat{\sigma}^2_{X_{[,1:10]}}$, because $\frac{1}{50-6} < \frac{1}{50-11}$ and $\sum_{i=1}^{50} (y_i - \hat{y}_i)^2 \approx \sum_{i=1}^{50} (y_i - \hat{y}_i)^2$.

Potential issues

- Issue 3: **Bias-variance tradeoff**. As a model is more complicated, the model fitness gets better. However, the prediction error with new (or future, test) data could be higher.



*Source: Machine Learning for *De Novo* Peptide Identification (McDonnell, 2023)

1. Trade-off between bias and variance:

Complexity	Bias	Variance
Low	high (underfit)	low
High	low	high (overfit)

2. We want an ideal model that finds a balance between bias and variance, minimizing the prediction error.

Nested models

- Consider two nested models:

$$\begin{aligned}\text{Simple model}: &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon \\ \text{Complex model}: &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \epsilon\end{aligned}$$

  Testing $H_0 : \beta_3 = \beta_4 = 0$ can be done using an *F*-test.

- Note:

$$\begin{aligned}\text{Simple model}: &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon \\ \text{Complex model}: &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon\end{aligned}$$

  Testing $H_0 : \beta_3 = 0$ can be done using an *t*-test or *F*-test, because these two tests are mathematically equivalent ($t_{df}^2 = F_{1,df}$).

Nested models with a higher complexity term

- **Note**: Removing a lower complexity term while keeping a higher complexity term could result in unexpected consequences and is often not recommended.
- Case 1: Consider a quadratic regression model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i$$

  1. Suppose $H_0 : \beta_1 = 0$ is not rejected, but $H_0 : \beta_2 = 0$ is rejected.
  2. If we remove the nonsignificant linear term $\beta_1 x_i$, the model becomes $y_i = \beta_0 + \beta_2 x_i^2$, which is symmetric about $x = 0$. Therefore, if $\hat{\beta}_2$ is significant, we often do not remove $\beta_1 x_i$.

- Case 2: Consider a two-way ANOVA model

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

  1. Suppose $H_0 : \beta_2 = \cdots = \beta_J = 0$ is not rejected, but $H_0 : (\alpha\beta)_{ij}$'s $= 0$ is rejected.
  2. If the interaction effects are significant, then the main effects exist. Therefore, we do not remove main effects if interaction effects are significant.

Testing-based model selection

- Suppose a data set contains observations of the response variable Y and p predictors $X_1, \ldots, X_p$ from a sample of n subjects.
- How do we find a **best (optimal) subset** of the p predictors for our linear model?
- Given a set of p predictors, an exhaustive search involves 2^p different combinations for predictor selection, resulting in 2^p different model fits. (Note: $2^{30} \approx 1.074$ billion)
- Testing-based model selection methods:
  1. Backward selection
  2. Forward selection
  3. Stepwise selection

Backward selection

- **Backward selection** procedures
  1. Start with a likely too complex model.
  2. Remove each predictor and compare this reduced model to the complex model by p-value from t-test (or F-test).
  3. Remove the predictor with the largest p-value from the complex model. This reduced model becomes a new complex model.
  4. Repeat steps 2 and 3 until all remaining predictors have significant p-values.

- Notes
  1. The cut-off value for significance in each step does not need to be $\alpha = 0.05$.
  2. We often choose a larger significance level if prediction performance is important.

Forward selection

- **Forward selection** procedures
  1. Start with a likely too simple model.
  2. Include each predictor and compare the new model to the simple model by p-value from t-test (or F-test).
  3. Include the predictor with the smallest p-value to the simple model. This model becomes a new simple model.
  4. Repeat steps 2 and 3 until all included predictors have significant p-values. (Stop if every included predictor is no longer significant.)

Stepwise selection

- **Caveats** in backward and forward selection methods
  1. Once a predictor was removed in backward selection, the predictor will never be included in the model.
  2. Once a predictor was included in forward selection, the predictor will never be removed from the model.

- **Stepwise selection**
  1. At every stage, a predictor is included or removed.
  2. Stepwise selection is most commonly used in practices.
  3. However, no rigorous theory exist if stepwise selection is always better than backward and forward selections.

Common issues

- The three selection methods have some common issues.
  1. Because of "one-at-a-time" addition/deletion mechanism, the selected model might not be the most optimal model.
  2. The methods ignore the science behind the data, possibly ignoring predictors that are scientifically meaningful, although there are ways to lessen this caveat.
  3. Some predictors in the selected model could be highly co-linear, exhibiting the multicollinearity issue.

Information criteria | Motivation

- Consider two models:

$$\text{Model 1} : y = \beta_0 + \beta_1 x_1 + \epsilon$$
$$\text{Model 2} : y = \beta_0 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$$

  F-tests cannot compare these two models, because the models are not nested.

- **Question**: How do we compare these two models?
- Recall the bias-variance tradeoff!
  1. If a model is too simple, the model has high bias (underfit).
  2. As a model becomes more complicated, the model tends to have low bias.
  3. However, a more complicated model tends to have higher variance (overfit).

Information criteria | Idea

- **Information Criterion (IC)** approach is often used to compare non-nested models.
- Most information criteria consist of two parts:

$$\text{IC} = \text{Model fitness} + \text{Model penalty}$$

1. As a model becomes more complicated, the model fitness term becomes smaller.
2. The model penalty term penalizes complicated models, producing a larger value for a more complicated model.
3. IC finds an optimal model that balances model fit and model complexity — a not-too-simple, not-too-complicated model with good accuracy in prediction.
4. A model with a **smaller** IC is considered a **better** model.

- Examples: AIC, AICC, BIC, their variants, and so on.

Likelihood function & Maximum likelihood estimation

- The model fitness in IC is often based on the likelihood (or negative log-likelihood) function.
- Loosely speaking, a **likelihood function**, $L(\text{parameters} \mid \text{data})$, is a real-valued function defined on a parameter space that returns the probability of observing a particular data across different parameter values.
- **Maximum likelihood estimates (MLEs)** are the values of the parameters that maximize the likelihood function (or minimize the negative log-likelihood function).

Maximum likelihood estimation | A simple example

- Example 9.1. Suppose a box contains three balls. We know each of the balls may be orange or blue, but we do not know the total number of each color. Two balls are to be randomly selected *without* replacement. If two orange balls are selected ($X = 2$), what would be a good estimate of the total number θ of orange balls in the box?
  1. With two orange balls selected ($X = 2$), there are two possible cases: (i) two orange balls and one blue ball ($\theta = 2$) or (ii) three orange balls ($\theta = 3$).
  2. If case (i) is true ($\theta = 2$), the probability of randomly selecting two orange balls is $\frac{\binom{2}{2}\binom{1}{0}}{\binom{3}{2}} = \frac{1}{3}$.
  3. If case (ii) is true ($\theta = 3$), the probability of randomly selecting two orange balls is 1.
  4. The likelihood function for the number θ of orange balls is
  $$L(\theta|X = 2) = \begin{cases} \frac{1}{3}, & \text{if } \theta = 2; \\ 1, & \text{if } \theta = 3 \end{cases}$$
  5. **Question**. What is your estimate of the number θ of orange balls in the box? Is it two or three?
  6. **Note**: The probability of obtaining the observed sample is maximized when $\theta = 3$. Therefore, the MLE is $\hat{\theta} = 3$.

AIC

- Suppose we observed the data and fitted a linear model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ with $\varepsilon \sim \mathrm{N}(\mathbf{0}, \sigma^2\mathbf{I})$.
- **Akaike Information Criterion (AIC)** is

$$\mathrm{AIC} = -2\log L(\hat{\boldsymbol{\beta}}, \hat{\sigma}^2|\mathbf{X}, \mathbf{y}) + 2k$$

1. The first term, $-2\log L(\hat{\boldsymbol{\beta}}, \hat{\sigma}^2|\mathbf{X}, \mathbf{y})$, measures how well a fitted model explains observed data via a negative log-likelihood and can be reexpressed as

$$-2\log L(\hat{\boldsymbol{\beta}}, \hat{\sigma}^2|\mathbf{X}, \mathbf{y}) = \text{a constant} + n\log\hat{\sigma}^2$$

2. As a model fits the data better, the first term of AIC gets smaller.
3. In the second term, k denotes the number of the parameters in the fitted model. If a fitted model contains an intercept and p predictors, then $k = p + 2$. (Why? $\beta_0, \beta_1, \ldots, \beta_p, \sigma^2$).
4. The second term is a penalty. As a model is more complicated, the model is more penalized.
5. **Note**: The lower AIC is, the better the fitted model is.

BIC

- **Bayesian Information Criterion (BIC)** is

$$\text{BIC} = -2 \log L(\hat{\boldsymbol{\beta}}, \hat{\sigma}^2 | \mathbf{X}, \mathbf{y}) + k \log n$$

  1. The first term of BIC is the same as that of AIC.
  2. The second term, $k \log n$, penalizes more heavily a more complicated model than AIC, because $\log n > 2$ (or $n > e^2 \approx 7.389$) in most studies.
  3. **Note**: The lower BIC is, the better the fitted model is.

- Notes
  1. AIC and BIC can be used to compare models that are not nested.
  2. AIC and BIC should be used for the models with the same response variable.

Adjusted R^2

- R^2 is calculated by

$$R^2 = \frac{SSR_{simple} - SSR_{complex}}{SSR_{simple}} = 1 - \frac{SSR_{complex}}{SSR_{simple}}$$

  1. SSR_{simple} is the SSR of the intercept only model ($y = \beta_0 + \epsilon$).
  2. $0 < R^2 < 1$. A model with a larger value of R^2 is preferred.
  3. Caveat: However, R^2 prefers more complicated models, because as p^* increases, R^2 also increases.

- To overcome this issue, we use an **adjusted R^2**:

$$R^2_{adj} = 1 - \frac{SSR_{complex}/(n-p^*)}{SSR_{simple}/(n-1)} = 1 - \frac{MSR_{complex}}{MSR_{simple}} = 1 - \frac{\hat{\sigma}^2_{complex}}{\hat{\sigma}^2_{simple}}$$

  1. As p^* increases, R^2_{adj} does not necessarily increase.
  2. Adding a predictor to the model that has no predictive power will decrease R^2_{adj}.
  3. **Note**: The closer to 1 R^2_{adj} is, the better the fitted model is.

Summary

- Variable selection methods
  1. Backward selection
  2. Forward selection
  3. Stepwise selection
  4. Lasso, Random Forest: STA 578
- Model comparison methods
  1. t-test, F-test
  2. Information criteria: AIC, AICC, BIC
  3. Adjusted R^2
  4. Likelihood Ratio (LR) test: STA 574 & STA 674, Cross-Validation (CV): STA 578
- Another approach – Dimension reduction
  1. Principal Component Analysis (PCA): STA 572
  2. Factor Analysis (FA): STA 572
  3. Linear Discriminant Analysis (LDA): STA 572
  4. Autoencoders: MAT 599(?)